Review plan · for the teacher

Teaching decisions: unit check

Optimization · Level 4 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- Two goals at once
- Fair by design
- When the model is wrong
- Big problems, good answers

Activity: Decision clinic

1. Warm-up (5 min): Each teacher names one school decision that people complain about: the timetable, duties, exam week or the bus.
2. Quiz (15 min): Teams of four read each question aloud, agree one answer, then hear the explanation. Write the hardest question on the board.
3. Clinic (15 min): Each team takes one decision from the warm-up and fills four boxes: the options nobody beats, the fairness rule, the missing rules people would spot, and what good enough means.
4. Share (5 min): Each team names one change it will propose to the head teacher, and who must agree it.

Works as a staff training session as well as a class review: use real school decisions, and keep names out of the examples.

Discussion

- Which idea from this unit will you use first?
- Where does our school hide a trade-off inside a single number?
- Who should be in the room when a fairness rule is chosen?

Exit ticket

- One decision I will lead differently, and the first step I will take.

Try it at school tomorrow

Pick one decision at your school this month and run it the way this unit suggests: cross out the beaten options, agree the weights and a fairness rule before anyone sees the winner, test the plan with the people it affects, and publish how it was made.

Quiz sheet · for each team

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Team _____ Date _____

1 **Three bus routes: X (30 km, longest ride 50 min), Y (32 km, 45 min) and Z (33 km, 50 min). Which route is beaten?**

◯ Y ◯ Z ◯ X ◯ None of them

2 **The weights for combining two goals are facts that the person building the model should work out alone.**

◯ True ◯ False

3 **Which of these are signs that a model is missing a rule? Choose all that apply.**

☐ Last year's real timetable breaks one of the model's rules

☐ The plan barely changes when a price changes

☐ The cheapest lunch serves a spoonful of beans

☐ The cooks say the plan can't be cooked in time

☐ The solver reports OPTIMAL

4 **Match each idea to what it means.**

Pareto front	• • • •	What one more unit of a limit is worth
Shadow price	• • • •	A plan no single small change improves
Price of fairness	• • • •	The plans no other plan beats
Local best	• • • •	What the total gives up for the worst-off

5 **What does this program print?**

```
plans = [("A", 4, 9), ("B", 5, 4),
         ("D", 6, 1)]
w = 4  # a day counts as 4 pupils
def score(p):
    return w * p[1] + p[2]
best = min(plans, key=score)
print(best[0])
```

◯ B ◯ A ◯ D

6 **Complete the sentences.**

Solve again with a limit one unit larger: the change in the best goal value is that limit's _____. A limit with room to spare has a value of _____.

Word bank: zero · gap · shadow price · weight

7 **Complete the model so it makes the worst-off teacher as well off as possible.**

```
worst = m.new_int_var(1, 9, "worst")
for t in range(3):
    m.add(score[t] _____ worst)
m. _____ ( _____ )
```

Word bank: `maximize` `>=` `total` `minimize` `<=` `worst`

8 **Put in order: fixing a model that gave a plan people can't use.**

- [ ] Sort their objections into missing rules and wrong numbers
- [ ] Write the assumptions beside the new plan
- [ ] Show the plan to the people who live with it
- [ ] Add the rules, correct the numbers and solve again

9 **Simulated annealing sometimes keeps a swap that makes the plan worse, and that can lead it to a better plan in the end.**

◯ True ◯ False

10 **This program should keep the plans nobody beats, but the front comes out empty. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  front = []
2  for name, days, clash in plans:
3      beaten = False
4      for other, d2, c2 in plans:
5          if d2 <= days and c2 <= clash:
6              beaten = True
7      if not beaten:
8          front.append(name)
```

Answers · for the teacher

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1 **Z**

X is shorter than Z with the same longest ride, so Z is beaten (Y beats it too).

2 **False**

Weights are values: how much one goal matters against another.

3 **The cooks say the plan can't be cooked in time; The cheapest lunch serves a spoonful of beans; Last year's real timetable breaks one of the model's rules**

Cooking time, whole portions and last year's exception are rules the world has and the model doesn't.

4 **Pareto front — The plans no other plan beats**
Shadow price — What one more unit of a limit is worth
Price of fairness — What the total gives up for the worst-off
Local best — A plan no single small change improves

These four ideas let you show a decision honestly: which plans are worth considering, what a limit is worth, what fairness costs, and how sure a search can be.

5 **B**

With a day counting as 4 pupils, the scores are A 4 × 4 + 9 = 25, B 4 × 5 + 4 = 24 and D 4 × 6 + 1 = 25.

6 **Solve again with a limit one unit larger: the change in the best goal value is that limit's shadow price. A limit with room to spare has a value of zero.**

In the garden, one more pupil-hour raises the best profit from 290 to 295 (worth 5), while extra water is worth nothing because 3 cans are already spare.

7

```
worst = m.new_int_var(1, 9, "worst")
for t in range(3):
    m.add(score[t] >= worst)
m.maximize(worst)
```

Every teacher's score must be at least worst, and the goal pushes worst as high as it can go, so the lowest score is lifted.

8 **Top to bottom: 2, 4, 1, 3**

People find what the model missed; sorting tells you whether to add a limit or fix a number; then you solve again, and the written assumptions let the next person check the new plan.

9 **True**

It keeps some worse swaps on purpose, often while the temperature is high and rarely as it cools.

10 **Line 5:** `if d2 <= days and c2 <= clash and (d2 < days or c2 < clash):`

Every plan is no worse than itself on both goals, so without "and better on at least one", each plan beats itself and nothing is kept.